

Sandesh S. Kalantre

Knight-Hennessy Fellow, Stanford University | [Website](#) | sandykal@stanford.edu

EDUCATION

Stanford University

June 2022 - Sep 2026 (*exp.*)

PhD in Physics, Knight-Hennessy Fellow

Advisor: Prof. David Goldhaber-Gordon

University of Maryland (UMD), College Park

Aug 2018 - May 2022

JQI-QuICS Lanczos Graduate Fellow (transferred to Stanford)

Advisor: Prof. James R. Williams

Indian Institute of Technology (IIT), Bombay - India

Aug 2014- May 2018

B.Tech in Engineering Physics with Honors and Minor in CS, GPA: 9.87/10

HONORS AND AWARDS

- Knight-Hennessy Fellowship (\$60,000/yr) 2023-2026
For visionary & collaborative leaders addressing complex real-world challenges at Stanford
- Lanczos Graduate Fellowship 2018-2020
Awarded to top incoming 2-3 students (\$36,000/yr) at Joint Quantum Institute, UMD
- Institute Silver Medal for graduating at the top of Engineering Physics program, IIT Bombay 2018
- S. N. Bose Scholarship (\$2000) awarded to top 50 students from India in STEM 2017
- Institute Academic Award and R.P. Singh Memorial Award, IIT Bombay 2015-2017
- ISTernship at IST Austria (\$2500) awarded to around 40 students worldwide 2016
- National Initiative on Undergraduate Science (NIUS) Program in Physics 2014-2015
- Kishore Vigyan Protsahan Yojana (KVPY), financial award for basic science high-school students 2013
- 2 Gold and 1 Silver medals at International Olympiads on Astronomy and Astrophysics 2011-2013
- National Talent Search Scholarship (NTSE) in science and math for high school students 2010

PATENTS

- Ray-based Classification Framework for Machine Learning-based Tuning Techniques 2023
J.P. Zwolak, **S.S. Kalantre**, J.M. Taylor, T.W. McJunkin, US Patent App. 18/028,154

PUBLICATIONS

500+ citations, h-index = 10 ([Google Scholar](#)) — 12 publications

13. **S.S. Kalantre**, J. Nordlander, M.A. Anderson, J.A. Mundy, and David Goldhaber-Gordon, Microwave spin resonance in epitaxial thin films of spin liquid candidate TbInO₃ arXiv:2603.14545, 2026
12. D.L. Buterakos, **S.S. Kalantre**, J. Ziegler, J.M. Taylor, J.P. Zwolak, QDFlow: A Python package for physics simulations of quantum dot devices, arXiv preprint arXiv:2509.13298, 2025
11. M. Pendharkar, S.J. Tran, G. Zaborski Jr, J. Finney, A.L. Sharpe, R.V. Kamat, **S.S. Kalantre**, et. al. and Marc A. Kastner, Andrew J. Mannix, David Goldhaber-Gordon, Torsional force microscopy of van der Waals moirés and atomic lattices, *Proceedings of the National Academy of Sciences* 121 (10), e2314083121, 2024
10. J. Ziegler, T. McJunkin, E.S. Joseph, **S.S. Kalantre**, B. Harpt, D.E. Savage, et. al. and M.A. Eriksson, Jacob M. Taylor, Justyna P Zwolak, Toward robust autotuning of noisy quantum dot devices, *Physical Review Applied* 17 (2), 024069, 2022

9. B.J. Weber, **S.S. Kalantre**, T. McJunkin, J.M. Taylor, J.P. Zwolak, Theoretical bounds on data requirements for the ray-based classification, *SN Computer Science* 3 (1), 57, 2022
8. C.J. Trimble, M.T. Wei, N.F.Q. Yuan, **S.S. Kalantre** et. al. and J.R. Williams, Josephson Detection of Time Reversal Symmetry Breaking Superconductivity in SnTe Nanowires, *Nature Partner Journal (npj), Quantum Materials* 6 (1), 1-6, 2021
7. **S.S. Kalantre**, F. Yu, M.T. Wei, et. al. and J.R. Williams, Anomalous Phase Dynamics of Driven Graphene Josephson junctions, *Physical Review Research* 2 (2), 023093, 2020
6. J.P. Zwolak, T. McJunkin, **S.S. Kalantre**, J.P. Dodson, E.R. MacQuarrie, et. al. and M.A. Eriksson, J.M. Taylor, Autotuning of Double-Dot Devices In Situ with Machine Learning, *Physical Review Applied* 13 (3), 034075, 2020
5. J.P. Zwolak, T. McJunkin, **S.S. Kalantre**, J.P. Dodson, E.R. MacQuarrie, et. al. and M.A. Eriksson, J.M. Taylor, Ray-based framework for state identification in quantum dot devices, *PRX Quantum* 2 (2), 020335, 2020
4. J.P. Zwolak, **S.S. Kalantre**, T. McJunkin, B.J. Weber, J.M. Taylor, Ray-based classification framework for high-dimensional data, arXiv preprint arXiv:2010.00500, Proceedings of the Machine Learning and the Physical Sciences Workshop at *NeurIPS 2020*, Vancouver, Canada
3. **S.S. Kalantre**, J.P. Zwolak, et. al. and J.M. Taylor, Machine learning techniques for state recognition and auto-tuning in quantum dots, *Nature Partner Journal (npj), Quantum Information* 5 (1), 1-10, 2019
2. P. Sriram, **S.S. Kalantre**, K. Gharavi, J. Baugh, B. Muralidharan, Supercurrent interference in semiconductor nanowire Josephson junctions, *Physical Review B* 100 (15), 155431, 2019
1. J.P. Zwolak, **S.S. Kalantre**, X. Wu, S. Ragole, J.M. Taylor, QFlow lite dataset: A machine-learning approach to the charge states in quantum dot experiments, *PloS one* 13 (10), e0205844, 2018

INVITED CONFERENCE PRESENTATIONS

1. Microwave resonance in thin films of spin liquid candidate TbInO₃, 7th International Workshop on Complex Oxides, Cargese, France, March 2024

CONTRIBUTED CONFERENCE PRESENTATIONS

1. Hybrid superconducting circuit architecture to probe van der Waals heterostructures, APS March Meeting, March 2026
2. Microwave resonance probes of frustrated magnets, Capri School on Physics of Superconducting Devices, Capri, Italy, April 2025
3. Flip-chip spectroscopy of frustrated magnet Gd₂Ti₂O₇, MURI program on Interfacial Superconductivity, Annual Review, Stanford University, Dec 2024
4. Low temperature magneto-transport of epitaxial samarium hexaboride films, APS March Meeting, March 2024
5. Low temperature magnetotransport of SmB₆ films, MURI program on Interfacial Superconductivity, Annual Review, Harvard University, Dec 2023
6. Electron spin resonance measurements of spin liquid candidate TbInO₃ with superconducting resonators, APS March Meeting, March 2023
7. Electron spin resonance measurements of spin liquid candidate TbInO₃ with superconducting resonators, MURI program on Interfacial Superconductivity, Annual Review, Harvard University, Dec 2022

8. Electron Spin Resonance near the Metal Insulator Transition in Phosphorous doped Si, APS March Meeting, APS March Meeting, March 2022
9. Anomalous Phase Dynamics of graphene Josephson Junctions, APS March Meeting, APS March Meeting, March 2020
10. In-situ machine learning assisted auto-tuning of quantum dot devices, NISQ, UMD, June 2019
11. Machine Learning Techniques for quantum dots, JQI-QuICS-CMTC seminar, UMD, Dec 2018
12. Non-linear dynamics in graphene Josephson junctions, APS MAS, UMD, Nov 2018
13. ML and Quantum Dots, APS March Meeting, Los Angeles, CA, March 2018
14. ML and Quantum Dots, Quantum Seminar, NIST and UMD, July 2017

RESEARCH

Geballe Lab for Advanced Materials, Stanford University

June 2022 - Present

Advisor: Prof. David Goldhaber-Gordon

- Engineered a scalable NbTiN superconducting resonator platform with flip-chip sample integration for high-throughput spin resonance characterization, with quick sample turnaround time for novel frustrated magnets and quantum spin liquid candidates.
- Designed and validated a cQED microwave architecture for superfluid stiffness extraction in van der Waals superconductors, demonstrating GHz-frequency materials characterization at mK temperatures.
- Quantified topological transport signatures in SmB_6 thin films and $\text{Eu}_3\text{In}_2\text{As}_4$ nanowires, building a systematic magneto-transport characterization pipeline for topological material candidates.

Quantum Materials Lab (QMDLab), University of Maryland

August 2018 - June 2022

Advisor: Prof. James R. Williams

- Characterized superconducting interfaces in topological materials and van der Waals heterostructures through multi-modal electrical characterization, combining quasi-DC transport and microwave Shapiro response measurements.
- Built an analytical modeling framework for Josephson junction dynamics under microwave drive, leveraging non-linear oscillator theory to extract device parameters and predict Shapiro step signatures.

National Institute for Standards and Technology - Gaithersburg

May 2017 - July 2017

Advisor: Prof. Jacob M. Taylor

- Developed CNN-based classification models for automated charge state recognition in semiconductor quantum dots, directly enabling downstream closed-loop device control.
- Engineered one of the first auto-tuning algorithms for quantum dots, autonomously driving an in-situ device to the single-electron regime — a first-of-its-kind demonstration in the field.

Indian Institute of Technology, Bombay

August 2016 - May 2018

Advisor: Prof. Suddhasatta Mahapatra

- Nano-fabrication of nano-scale devices in semiconductor heterostructures (SiGe)
- Worked on a theoretical problem of Electron Dipole Spin Resonance (EDSR) selectivity of Phosphorous donor qubits in micromagnet fields

Institute for Science and Technology (IST) - Austria, Vienna

May 2016 - July 2016

Advisor: Dr. Giorgos Katsaros

- Studied theoretical aspects of transport in quantum dot and nanowire systems in semi-conductors
- Worked on conductance measurements to observe Coulomb blockade and diamonds in p-type transistors at 4 K cooled with liquid He

- Non-equilibrium Green's function (NEGF) formalism to model electron transport in superconductor-semiconductor systems
- Determined the effects of angular momentum subbands and Andreev Bound states to the current-phase relationship in nanowires connected to superconducting contacts

RESEARCH SKILLS

Quantum Device Fabrication

- Thin film deposition: magnetron sputtering, e-beam evaporation, and atomic layer deposition (ALD) of superconducting and dielectric films.
- Lithography: photo and e-beam lithography (EBL) for superconducting resonator circuits, Josephson junctions, vdW heterostructures, and mesoscopic devices.
- Etching and surface processing: reactive ion etching (RIE), plasma and chemical etching, SEM imaging, and 2D material dry stacking.

Cryogenic and Microwave Instrumentation

- Cryogenic systems: mK dilution refrigerators, variable temperature inserts, and liquid He dipsticks for sub-Kelvin transport and microwave experiments.
- Microwave characterization: VNA-based spectroscopy, RF reflectometry, custom mixer circuits, and high-speed digitizers for pulsed cQED measurements.
- DC and low-frequency: lock-in amplifiers, quasi-DC transport, and SQUID magnetometry for magneto-transport and susceptibility measurements.

Simulation and Computational Tools

- Scientific computing: NumPy, SciPy, and QuTiP for modeling quantum dots, Josephson junctions, and 2D material devices; TensorFlow for ML-driven experimental automation.
- EM and multiphysics simulation: Sonnet, HFSS, and Ansys Maxwell for microwave circuit design; COMSOL for electrostatic and magnetostatic modeling.
- CAD and PCB design: SolidWorks, FreeCAD, and KiCad for experimental hardware design; metal machining and 3D printing for custom apparatus; automated CAD design with gdspy/gdstk

Communication and Leadership

- Grant and technical report writing; first-author publications in peer-reviewed journals.
- Mentorship of junior students, assistance in project management of multi-investigator collaborations

TEACHING EXPERIENCE

Teaching Assistant, Stanford University

2023-2025

- TA for undergraduate electricity and magnetism and quantum mechanics II
- Mentored student teams in the design of a nuclear quadrupole resonance setup & cryogenic probes for advanced undergraduate physics lab

Teaching Assistant, IIT Bombay

2015-2018

- TA for a range of physics & mathematics undergraduate courses on electricity and magnetism, introductory quantum physics and partial-differential equations; on two occasions was also the head TA responsible for course logistics and personnel management

LEADERSHIP EXPERIENCE

Organizational Volunteer and Board Member, USAAAO

2018-Present

- Designed astronomy and astrophysics problems at the high-school level, and assisted in coaching the US national team to the International Olympiad on Astronomy and Astrophysics (IOAA)

General Secretary, Department of Physics, IIT Bombay

2017-2018

- Elected as the top student representative of the physics department's bachelor and masters students who communicated between students, faculty and other representatives at the institute level; work led to curriculum changes and bridged the gap in student-faculty communication in the department

Institute Student Mentor, IIT Bombay

2017-2018

- Senior mentor to a wing of 12 freshmen to assist them in settling to life at a university

Academic Resource Person, International Physics Olympiad (IPhO), Mumbai

2015

- Exam grader; worked in a team of 2 in development of IPho-rum — a browser based application for tasks such as voting, translation upload and feedback submission among ≈ 100 users

MISCELLANEOUS INTERESTS

- Science popularization at IIT Bombay at the undergraduate-level, gave talks on varied topics such as Celestial mechanics and N-body problem, Cosmology and Quantum Computing
- Violin and classical music
- Won a Bronze medal in Astronomy hackathon as part of InterIIT Tech Meet
- Secured first place with a team of five in *Online Physics Brawl 2016* in the open category
- Teaching volunteer at NGOs for underprivileged children in Powai, Mumbai